ECON1050 Tutorial 5

Calculus II: Derivatives in Use

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Today's Plan

- Recap: key ideas from Lecture 5
- Tutorial questions: Q1–Q5

Recap: Explicit and Implicit Functions

Explicit function

An explicit function writes the dependent variable directly as a function of the independent variable:

$$y = f(x).$$

Implicit function

An implicit relationship is written in the form

$$g(x, y) = c,$$

where y is understood to be a function of x even though it has not been isolated.

Recap: Implicit Differentiation

Procedure

To differentiate an implicitly defined function:

1. Differentiate both sides with respect to x.
2. Treat y as a function of x and apply the chain rule whenever differentiating a term involving y.
3. Use product or quotient rules when required.
4. Solve the resulting equation for $y' = \frac{dy}{dx}$.

Recap: Higher-Order Implicit Derivatives

Second derivative

Once an equation for y' has been obtained, differentiate again with respect to x to find

$$y'' = \frac{\mathrm{d}^2 y}{\mathrm{d}x^2}.$$

Terms containing y or y' must again be differentiated using the chain and product rules as appropriate.

Evaluation at a point

After finding the derivative expression, substitute the specified values of x, y, and, where necessary, y'.

Recap: Linear Approximation

Linear approximation around $x = a$

For x close to a,

$$f(x) \approx f(a) + f'(a)(x - a).$$

Interpretation

The linear approximation is the tangent line to f at $x = a$. Therefore, the function and its approximation have the same value and the same first derivative at a.

Recap: Quadratic Approximation

Quadratic approximation around $x = a$

If f is twice differentiable,

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2.$$

Matching conditions

At $x = a$, the quadratic approximation matches $f(x)$ in value, first derivative, and second derivative.

Recap: Taylor Polynomial

nth-order approximation around $x = a$

$$f(x) \approx f(a) + \sum_{i=1}^{n} \frac{f^{(i)}(a)}{i!}(x-a)^i.$$

Equivalently,

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Recap: Maclaurin Polynomial and Remainder

Maclaurin polynomial

When the expansion point is $a = 0$, the Taylor polynomial becomes

$$
\begin{aligned}
f(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + R_{n+1}(x) \\
&\approx f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.
\end{aligned}
$$

Remainder

Taylor's formula adds a remainder term $R_{n+1}(x)$, which measures the approximation error.

Recap: Elasticity

General definition

For a differentiable function $f(x)$,

$$\mathrm{El}_x f(x) = \frac{xf'(x)}{f(x)}.$$

Elasticity measures the percentage responsiveness of $f(x)$ to a percentage change in x.

Demand elasticity

For a demand function $Q(p)$,

$$\varepsilon(p) = \frac{pQ'(p)}{Q(p)}.$$

Recap: Elastic, Unit Elastic and Inelastic

Classification

$$
\begin{aligned}
|\operatorname{El}_x f(x)| &> 1 && \text{elastic,} \\
|\operatorname{El}_x f(x)| &= 1 && \text{unit elastic,} \\
|\operatorname{El}_x f(x)| &< 1 && \text{inelastic,} \\
|\operatorname{El}_x f(x)| &= 0 && \text{perfectly inelastic,} \\
|\operatorname{El}_x f(x)| &= \infty && \text{perfectly elastic.}
\end{aligned}
$$

Recap: Elasticity and Revenue

Revenue

If demand is $D(p)$, revenue is

$$R(p) = pD(p).$$

The elasticity of revenue satisfies

$$\mathrm{El}_p\, R(p) = 1 + \mathrm{El}_p\, D(p).$$

Implication for price changes

The marginal revenue generated by a price change is positive when demand is inelastic and negative when demand is elastic.

Tutorial Q1

Question 1

1. Given y is a differentiable function of x, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ by implicit differentiation when
$$x - y + 3xy = 2.$$

2. Assume that y is a function of x. Find y' for
$$x^3 + y^3 = 4.$$

Tutorial Q1.1: First Derivative

Knowledge used

- Implicit differentiation: differentiate both sides with respect to x.
- Product rule for the term xy.
- Treat y as a function of x, so $\frac{\mathrm{d}y}{\mathrm{d}x} = y'$.

Solution

From

$$x - y + 3xy = 2,$$

differentiate with respect to x:

$$1 - y' + 3y + 3xy' = 0.$$

Collect the y' terms:

$$y'(-1 + 3x) = -1 - 3y.$$

Therefore,

$$\boxed{y' = \frac{1 + 3y}{1 - 3x}}.$$

Tutorial Q1.1: Second Derivative

Knowledge used

- Differentiate the first-derivative equation again.
- Product rule for xy'.
- Substitute the previously obtained expression for y'.

Solution

Differentiate

$$1 - y' + 3y + 3xy' = 0$$

again:

$$-y'' + 3y' + 3y' + 3xy'' = 0.$$

Hence

$$y''(3x - 1) = -6y' \implies y'' = \frac{6y'}{1 - 3x}.$$

Substitute $y' = \frac{1+3y}{1-3x}$:

$$\boxed{y'' = \frac{6 + 18y}{(1 - 3x)^2}}.$$

Tutorial Q1.2: Implicit Differentiation

Knowledge used

- Chain rule for a power of an implicitly defined variable.
- $\frac{\mathrm{d}}{\mathrm{d}x}(y^3) = 3y^2 y'$.

Solution

From

$$x^3 + y^3 = 4,$$

differentiate with respect to x:

$$3x^2 + 3y^2 y' = 0.$$

Solve for y':

$$3y^2 y' = -3x^2 \quad \Longrightarrow \quad \boxed{y' = -\frac{x^2}{y^2}}.$$

Tutorial Q2

Question 2

Assume y is a differentiable function of x that satisfies

$$2x^2 + 6xy + y^2 = 18.$$

Find y' and y'' at the point

$$(x, y) = (1, 2).$$

Tutorial Q2: First Derivative

Knowledge used

- Implicit differentiation.
- Product rule for xy.
- Chain rule for y^2.

Solution

Differentiate:

$$4x + 6y + 6xy' + 2yy' = 0.$$

Thus

$$y'(6x + 2y) = -4x - 6y,$$

so

$$y' = -\frac{2x + 3y}{3x + y}.$$

At $(1, 2)$,

$$\boxed{y'(1,2) = -\frac{8}{5}}.$$

Tutorial Q2: Second Derivative

Knowledge used

- Quotient rule applied to the expression for y'.
- Evaluate the resulting expression at $(x, y) = (1, 2)$.

Solution

Starting from

$$y' = -\frac{2x + 3y}{3x + y},$$

the tutorial solution simplifies the second derivative to

$$y'' = \frac{42xy + 7y^2 + 14x^2}{(3x + y)^3}.$$

Therefore,

$$y''(1, 2) = \frac{42(1)(2) + 7(2)^2 + 14(1)^2}{(3 + 2)^3} = \boxed{\frac{126}{125}}.$$

Tutorial Q3

Question 3

Find the linear and quadratic approximations to

$$f(x) = (1 + x)^5$$

about $x = 0$, and use them to approximate

$$1.05^5.$$

Tutorial Q3: Linear Approximation

Knowledge used

- Linear approximation:
$$f(x) \approx f(a) + f'(a)(x-a).$$
- Here the expansion point is $a = 0$.

Solution

For
$$f(x) = (1+x)^5, \qquad f'(x) = 5(1+x)^4,$$
we have
$$f(0) = 1, \qquad f'(0) = 5.$$
Hence
$$\boxed{f(x) \approx 1 + 5x}.$$
With $x = 0.05$,
$$\boxed{1.05^5 \approx 1.25}.$$

Tutorial Q3: Quadratic Approximation

Knowledge used

- Quadratic approximation:
$$f(x) \approx f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2.$$
- Compare the linear and quadratic approximations at the same nearby value.

Solution

Since
$$f''(x) = 20(1+x)^3, \qquad f''(0) = 20,$$
we obtain
$$\boxed{f(x) \approx 1 + 5x + 10x^2}.$$
At $x = 0.05$,
$$1.05^5 \approx 1 + 5(0.05) + 10(0.05)^2 = \boxed{1.275}.$$
The calculator value is
$$1.05^5 = 1.276281563.$$

Tutorial Q4

Question 4

Let

$$f(x) = \sqrt{2x}.$$

Find the third-order Taylor polynomial of $f(x)$ about

$$a = 1.$$

Tutorial Q4: Derivatives at the Expansion Point

Knowledge used

- Third-order Taylor polynomial around a.
- Power rule and chain rule to obtain f', f'', and f'''.

Solution

Write

$$f(x) = (2x)^{1/2}.$$

Then

$$f'(x) = (2x)^{-1/2},$$

$$f''(x) = -(2x)^{-3/2},$$

$$f'''(x) = 3(2x)^{-5/2}.$$

At $a = 1$,

$$f(1) = \sqrt{2}, \quad f'(1) = \frac{1}{\sqrt{2}}, \quad f''(1) = -\frac{1}{\sqrt{2^3}}, \quad f'''(1) = \frac{3}{\sqrt{2^5}}.$$

Tutorial Q4: Third-Order Taylor Polynomial

Knowledge used

- Substitute derivative values into

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3.$$

Solution

For $a = 1$,

$$\boxed{\sqrt{2x} \approx \sqrt{2} + \frac{1}{\sqrt{2}}(x-1) - \frac{1}{2\sqrt{2^3}}(x-1)^2 + \frac{1}{2\sqrt{2^5}}(x-1)^3}.$$

Tutorial Q5

Question 5

For each demand function, calculate elasticity as a function of price and find the conditions under which demand is elastic.

1. $Q(p) = a - bp$, where $a > 0$ and $b > 0$.
2. $Q(p) = p^{-\lambda}$, where $\lambda > 0$.
3. $Q(p) = 100 - 4p^2$.

Use

$$\varepsilon(p) = \frac{pQ'(p)}{Q(p)}.$$

Tutorial Q5(a): Linear Demand

Knowledge used

- Elasticity: $\varepsilon(p) = \frac{pQ'(p)}{Q(p)}$.
- Following the tutorial solution, demand is elastic when $|\varepsilon(p)| \geq 1$.

Solution

For $Q(p) = a - bp$, $Q'(p) = -b$, so

$$\boxed{\varepsilon(p) = -\frac{bp}{a - bp}}.$$

Then

$$\left|-\frac{bp}{a - bp}\right| \geq 1 \quad \Longrightarrow \quad bp \geq a - bp \quad \Longrightarrow \quad \boxed{p \geq \frac{a}{2b}}.$$

Tutorial Q5(b): Constant Elasticity Demand

Knowledge used

- Apply the power rule to $Q(p) = p^{-\lambda}$.
- Substitute the derivative into the elasticity formula.

Solution

For

$$Q(p) = p^{-\lambda},$$

$$Q'(p) = -\lambda p^{-\lambda-1}.$$

Therefore,

$$\varepsilon(p) = \frac{p(-\lambda p^{-\lambda-1})}{p^{-\lambda}} = \boxed{-\lambda}.$$

Thus, following the tutorial solution, demand is elastic for all p if

$$\boxed{\lambda \geq 1}.$$

Tutorial Q5(c): Quadratic Demand

Knowledge used

- Use the power rule and the elasticity formula.
- Following the tutorial solution, solve $|\varepsilon(p)| \geq 1$.

Solution

For $Q(p) = 100 - 4p^2$, $Q'(p) = -8p$, hence

$$\boxed{\varepsilon(p) = -\frac{8p^2}{100 - 4p^2}}.$$

The elasticity condition gives

$$\left| -\frac{8p^2}{100 - 4p^2} \right| \geq 1 \implies 8p^2 \geq 100 - 4p^2 \implies p^2 \geq \frac{100}{12}.$$

Therefore,

$$\boxed{p \geq \frac{5}{\sqrt{3}}}.$$

Key Takeaways

- Implicit differentiation treats y as a function of x and uses the chain rule whenever y appears.
- Linear, quadratic and Taylor polynomials approximate a function locally using its derivatives at the expansion point.
- Elasticity measures percentage responsiveness and is computed as $\frac{xf'(x)}{f(x)}$.
- For demand, elasticity helps classify how strongly quantity responds to price changes.